



November 14, 2019

Cindy Tomaszewski
Cadillac WWTP
200 N. Lake St.
Cadillac, MI 49601

RE: Revision of report 358050 - 189760 for job 358050

Dear Cindy Tomaszewski:

Paragon Laboratories, Inc. received the samples associated with the workorder listed above for the analyses presented in the following report. The analyses pertain only to the aliquot of sample received.

This material is confidential and is intended solely for the person to whom it is addressed. If this is received in error, please contact the number below.

Please note that any unused portion of the sample(s) will be discarded 60 days after sample receipt, unless requested otherwise.

We appreciate the opportunity to assist you. If you have any questions concerning this report, please contact an Account Coordinator at (734) 462-3900.

Sincerely,

Margaret T Snyder
margarets@paragonlaboratories.com



SAMPLE SUMMARY

Workorder: 358050 CADILLACWWTP-102419

Lab ID	Sample ID	Sample Desc.	Matrix	Date Collected	Date Received	Collector
3580500001	Effluent	Grab	Wastewater	10/23/2019 07:30	10/24/2019 14:20	DS & CAT
3580500002	Effluent	Composite	Wastewater	10/23/2019 07:00	10/24/2019 14:20	DS & CAT



PROJECT SUMMARY

Workorder: 358050 CADILLACWWTP-102419

Workorder Comments

Report revised to include additional compounds not on original report.



ANALYTICAL RESULTS

Workorder: 358050 CADILLACWWTP-102419

Lab ID: 3580500001

Date Received: 10/24/2019 14:20 Matrix: Wastewater

Sample ID: Effluent

Date Collected: 10/23/2019 07:30

Sample Desc: Grab

PO: CAT191023PG

Parameters	Qualifier	Result Units	Reporting Limit	Dilution Factor	Result Qualifier		Analyzed	By
					Min	Max		
Volatiles								
Analysis Desc: EPA 624.1								
Acrolein		<5.0 µg/L	5.0	1			10/25/2019 16:06	DTM
Acrylonitrile		<1.0 µg/L	1.0	1			10/25/2019 16:06	DTM
Benzene		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
Bromodichloromethane		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
Bromoform		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
Bromomethane (Methyl bromide)		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
Carbon tetrachloride		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
Chlorobenzene		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
Chloroethane		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
2-Chloroethyl vinyl ether		<1.0 µg/L	1.0	1			10/25/2019 15:41	DTM
Chloroform		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
Chloromethane(Methyl chloride)		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
Dibromochloromethane		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
1,2-Dichlorobenzene		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
1,3-Dichlorobenzene		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
1,4-Dichlorobenzene		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
1,1-Dichloroethane		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
1,2-Dichloroethane (EDC)		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
1,1-Dichloroethene		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
trans-1,2-Dichloroethene		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
1,2-Dichloropropane		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
cis-1,3-Dichloropropene		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
trans-1,3-Dichloropropene		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
Ethylbenzene		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
Methylene chloride (DCM)		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
1,1,2,2-Tetrachloroethane		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
Tetrachloroethene (PCE)		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
Toluene		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
1,1,1-Trichloroethane		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
1,1,2-Trichloroethane		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
Trichloroethene (TCE)		<1.0 µg/L	1.0	1			10/25/2019 16:31	DTM
Vinyl chloride		<0.25 µg/L	0.25	1			10/25/2019 16:31	DTM

Individual Parameters

Analysis Desc: OIA-1677

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Report ID: 358050 - 200649

Revised Report

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ANALYTICAL RESULTS

Workorder: 358050 CADILLACWWTP-102419

Lab ID: **3580500001**

Date Received: 10/24/2019 14:20 Matrix: Wastewater

Sample ID: Effluent

Date Collected: 10/23/2019 07:30

Sample Desc: Grab

PO: CAT191023PG

Parameters	Qualifier	Result Units	Reporting Limit	Dilution Factor	Result Qualifier		Analyzed	By
					Min	Max		
Cyanide, Available		8.4 µg/L	2.0	1			10/30/2019 09:00	RLC
Analysis Desc: EPA 420.4								
Phenolics, Total		<0.050 mg/L	0.050	1			10/28/2019 07:30	GMT



ANALYTICAL RESULTS

Workorder: 358050 CADILLACWWTP-102419

Lab ID: **3580500002** Date Received: 10/24/2019 14:20 Matrix: Wastewater
 Sample ID: Effluent Date Collected: 10/23/2019 07:00
 Sample Desc: Composite PO: CAT191023PG

Parameters	Qualifier	Result Units	Reporting Limit	Dilution Factor	Result	Qualifier	Analyzed	By
					Min	Max		

Individual Parameters

Analysis Desc: EPA 605

Benzidine		<0.10 µg/L	0.10	1			10/31/2019 12:48	DTM
3,3'-Dichlorobenzidine		<1.5 µg/L	1.5	1			10/31/2019 12:48	DTM

Sample Preparation

Analysis Desc: EPA 605

Sep. Funnel Extraction EPA 605		Complete		1			10/29/2019 07:15	DTM
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Metals

Analysis Desc: EPA 200.8

Antimony, Total		<0.0010 mg/L	0.0010	1			10/30/2019 12:32	MKK
Arsenic, Total		0.076 mg/L	0.0010	1			10/30/2019 12:32	MKK
Barium, Total		0.015 mg/L	0.0050	1			10/30/2019 12:32	MKK
Beryllium, Total		0.0081 mg/L	0.00040	1			10/30/2019 12:32	MKK
Boron, Total		0.24 mg/L	0.0050	1			10/30/2019 12:32	MKK
Cadmium, Total		0.0095 mg/L	0.00020	1			10/30/2019 12:32	MKK
Chromium, Total		0.0069 mg/L	0.0010	1			10/30/2019 12:32	MKK
Lead, Total		<0.0010 mg/L	0.0010	1			10/30/2019 12:32	MKK
Silver, Total		0.0018 mg/L	0.00020	1			10/30/2019 12:32	MKK
Thallium, Total		<0.00040 mg/L	0.00040	1			10/30/2019 12:32	MKK
Zinc, Total		0.047 mg/L	0.0040	1			10/30/2019 12:32	MKK

Sample Preparation

Analysis Desc: EPA 625.1

Extraction for Semivolatiles		Complete		1			10/29/2019 12:00	DTM
Extraction for Semivolatiles		Complete		1			10/29/2019 12:00	DTM

Semivolatiles

Analysis Desc: EPA 625.1

Acenaphthene		<10 µg/L	10	1			11/1/2019 13:39	DTM
Acenaphthylene		<10 µg/L	10	1			11/1/2019 13:39	DTM
Anthracene		<10 µg/L	10	1			11/1/2019 13:39	DTM
Azobenzene (for 1,2-DPH)		<3.0 µg/L	3.0	1			11/1/2019 13:39	DTM
Benzo(a)anthracene		<10 µg/L	10	1			11/1/2019 13:39	DTM
Benzo(a)pyrene		<10 µg/L	10	1			11/1/2019 13:39	DTM
Benzo(b)fluoranthene		<10 µg/L	10	1			11/1/2019 13:39	DTM

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Revised Report

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ANALYTICAL RESULTS

Workorder: 358050 CADILLACWWTP-102419

Lab ID: 3580500002

Date Received: 10/24/2019 14:20 Matrix: Wastewater

Sample ID: Effluent

Date Collected: 10/23/2019 07:00

Sample Desc: Composite

PO: CAT191023PG

Parameters	Qualifier	Result Units	Reporting Limit	Dilution Factor	Result Qualifier		Analyzed	By
					Min	Max		
Benzo(ghi)perylene		<10 µg/L	10	1			11/1/2019 13:39	DTM
Benzo(k)fluoranthene		<10 µg/L	10	1			11/1/2019 13:39	DTM
Bis(2-chloroethoxy)methane		<10 µg/L	10	1			11/1/2019 13:39	DTM
Bis(2-chloroethyl)ether		<1.0 µg/L	1.0	1			11/1/2019 13:39	DTM
Bis(2-chloroisopropyl) ether		<10 µg/L	10	1			11/1/2019 13:39	DTM
Bis(2-ethylhexyl)phthalate		<10 µg/L	10	1			11/1/2019 13:39	DTM
4-Bromophenyl phenyl ether		<10 µg/L	10	1			11/1/2019 13:39	DTM
Butyl benzyl phthalate		<10 µg/L	10	1			11/1/2019 13:39	DTM
4-Chloro-3-methylphenol		<5.0 µg/L	5.0	1			11/1/2019 13:39	DTM
2-Chloronaphthalene		<10 µg/L	10	1			11/1/2019 13:39	DTM
2-Chlorophenol		<10 µg/L	10	1			11/1/2019 13:39	DTM
4-Chlorophenol		<10 µg/L	10	1			11/1/2019 13:39	DTM
4-Chlorophenyl phenyl ether		<10 µg/L	10	1			11/1/2019 13:39	DTM
Chrysene		<10 µg/L	10	1			11/1/2019 13:39	DTM
Di-n-butyl phthalate		<9.0 µg/L	9.0	1			11/1/2019 13:39	DTM
Di-n-octyl phthalate		<10 µg/L	10	1			11/1/2019 13:39	DTM
Dibenzo(ah)anthracene		<10 µg/L	10	1			11/1/2019 13:39	DTM
3,3'-Dichlorobenzidine		<10 µg/L	10	1			11/1/2019 13:39	DTM
2,4-Dichlorophenol		<10 µg/L	10	1			11/1/2019 13:39	DTM
Diethyl phthalate		<10 µg/L	10	1			11/1/2019 13:39	DTM
Dimethyl phthalate		<10 µg/L	10	1			11/1/2019 13:39	DTM
2,4-Dimethylphenol		<10 µg/L	10	1			11/1/2019 13:39	DTM
2,4-Dinitrophenol		<10 µg/L	10	1			11/1/2019 13:39	DTM
2,4-Dinitrotoluene		<10 µg/L	10	1			11/1/2019 13:39	DTM
2,6-Dinitrotoluene		<10 µg/L	10	1			11/1/2019 13:39	DTM
Fluoranthene		<1.0 µg/L	1.0	1			11/1/2019 13:39	DTM
Fluorene		<10 µg/L	10	1			11/1/2019 13:39	DTM
Hexachloroethane		<5.0 µg/L	5.0	1			11/1/2019 13:39	DTM
Indeno(1,2,3-cd)pyrene		<10 µg/L	10	1			11/1/2019 13:39	DTM
Isophorone		<10 µg/L	10	1			11/1/2019 13:39	DTM
2-Methyl-4,6-dinitrophenol		<10 µg/L	10	1			11/1/2019 13:39	DTM
Naphthalene		<10 µg/L	10	1			11/1/2019 13:39	DTM
Nitrobenzene		<10 µg/L	10	1			11/1/2019 13:39	DTM
2-Nitrophenol		<10 µg/L	10	1			11/1/2019 13:39	DTM
4-Nitrophenol		<10 µg/L	10	1			11/1/2019 13:39	DTM
N-Nitrosodi-n-propylamine		<10 µg/L	10	1			11/1/2019 13:39	DTM
N-Nitrosodimethylamine		<10 µg/L	10	1			11/1/2019 13:39	DTM
N-Nitrosodiphenylamine		<10 µg/L	10	1			11/1/2019 13:39	DTM

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ANALYTICAL RESULTS

Workorder: 358050 CADILLACWWTP-102419

Lab ID: **3580500002** Date Received: 10/24/2019 14:20 Matrix: Wastewater
 Sample ID: Effluent Date Collected: 10/23/2019 07:00
 Sample Desc: Composite PO: CAT191023PG

Parameters	Qualifier	Result Units	Reporting Limit	Dilution Factor	Result Qualifier		Analyzed	By
					Min	Max		
Pentachlorophenol		<1.8 µg/L	1.8	1			11/1/2019 13:39	DTM
Phenanthrene		<1.0 µg/L	1.0	1			11/1/2019 13:39	DTM
Phenol		<10 µg/L	10	1			11/1/2019 13:39	DTM
Pyrene		<10 µg/L	10	1			11/1/2019 13:39	DTM
2,3,7,8-TCDD (as TIC)	Not Detected	µg/L	10	1			11/1/2019 13:39	DTM
2,3,7,8-TCDD (as TIC)		<10 µg/L	10	1			11/1/2019 13:39	DTM
1,2,4-Trichlorobenzene		<10 µg/L	10	1			11/1/2019 13:39	DTM
2,4,6-Trichlorophenol		<5.0 µg/L	5.0	1			11/1/2019 13:39	DTM

Analysis Desc: EPA 612

Hexachlorobutadiene	<0.01 µg/L	0.01	1				10/30/2019 20:24	GFM
Hexachlorocyclopentadiene	<0.01 µg/L	0.01	1				10/30/2019 20:24	GFM
Hexachlorobenzene (HCB)	<0.01 µg/L	0.01	1				10/30/2019 20:24	GFM

Calculated Parameters

Analysis Desc: SM 2340 B

Hardness, Calcium as CaCO ₃	150 mg/L	1.0	1				10/30/2019 14:50	MKK
Hardness, Magnesium as CaCO ₃	56 mg/L	1.0	1				10/30/2019 14:50	MKK
Hardness, Total as CaCO ₃	200 mg/L	1.0	1				10/30/2019 14:50	MKK

Metals

Analysis Desc: EPA 200.7

Calcium, Total	59 mg/L	0.10	1				10/30/2019 14:50	MKK
Magnesium, Total	14 mg/L	0.10	1				10/30/2019 14:50	MKK



ANALYTICAL RESULTS QUALIFIERS

Workorder: 358050 CADILLACWWTP-102419

PARAMETER QUALIFIERS

Lab ID: 3580500002

- C 3-Methylphenol (m-Cresol) and 4-Methylphenol (p-Cresol) cannot be separated chromatographically, and are calibrated and reported as 4-Methylphenol (p-Cresol).